Lab Report – 07

Course Code: CSE 412

Course Title: Computer Graphics Lab

Experiment No: 07

Experiment Name: Implementation of 2D Transformation

Submitted To:

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Lab Report: 07

Implementation of 2D Transformation

Title: Drawing Four Colored Hexagons Using OpenGL (2D_Translation)

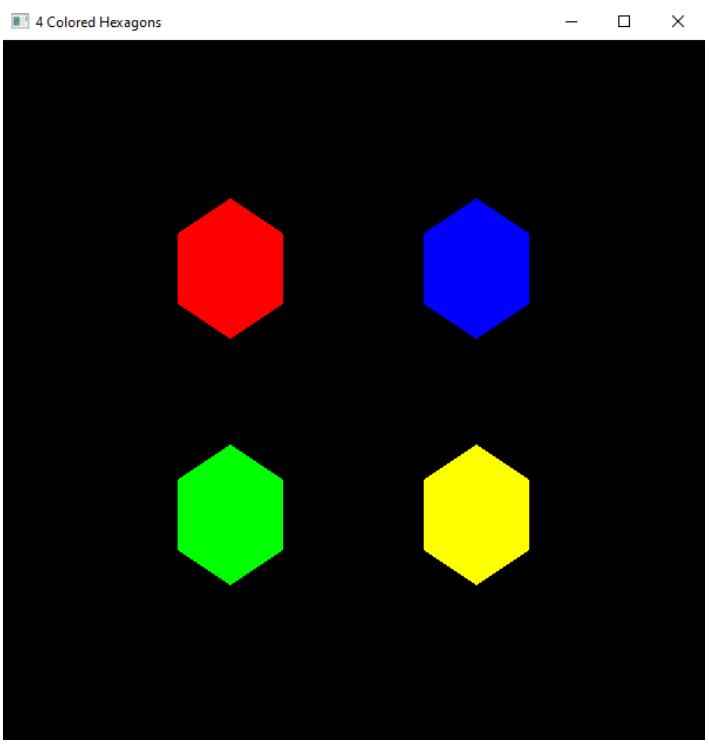
Introduction:

In this project, four hexagon shapes are drawn using OpenGL. Each hexagon has a different color and position.

Contents:

- **Functions Used:** drawHexagon(), glBegin(), glVertex2f(), glColor3f(), glTranslatef()
- **Shape Used:** Hexagon

Graph:



Draw X and Y axis and sketch four hexagons in four corners with different colors.

Code (Drawing Portion Only):

```
void drawHexagon()
{
    glBegin(GL_POLYGON);
    glVertex2f(0, 40);
    glVertex2f(30, 20);
    glVertex2f(30, -20);
    glVertex2f(0, -40);
    glVertex2f(-30, -20);
    glVertex2f(-30, 20);
    glEnd();
}
```

Discussion:

The task was to draw four hexagons. I used one hexagon function and placed it in four positions with different colors.

Title: Scaling of a Hexagon Using OpenGL

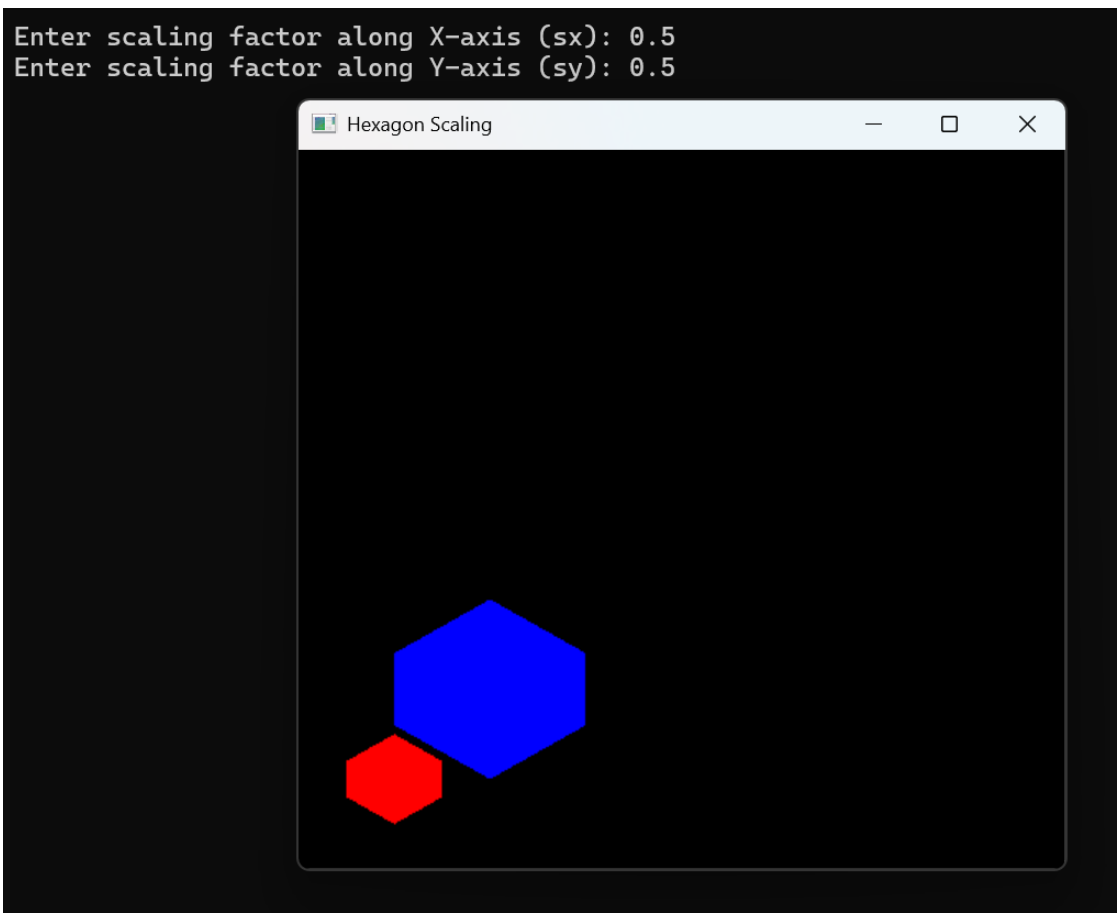
Introduction:

In this project, a hexagon is scaled using scaling transformation. Both original and scaled shapes are shown.

Contents:

- **Functions Used:** glBegin(), glVertex2f(), glColor3f()
- **Concept Used:** Scaling Transformation
- **Shape Used:** Hexagon

Graph:



Draw one original hexagon and another bigger/smaller hexagon based on scaling.

Code (Drawing Portion Only):

```
// Original Hexagon
glColor3f(0.0, 0.0, 1.0);
glBegin(GL_POLYGON);
for(int i = 0; i < 6; i++)
{
    glVertex2f(x[i], y[i]);
}
glEnd();

// Scaled Hexagon
glColor3f(1.0, 0.0, 0.0);
glBegin(GL_POLYGON);
for(int i = 0; i < 6; i++)
{
    glVertex2f(sx_h[i], sy_h[i]);
}
glEnd();
```

Discussion:

The task was to scale a hexagon. I multiplied each coordinate by scaling factors to create a new scaled shape.

Title: Translation of a Hexagon Using OpenGL (Rotation)

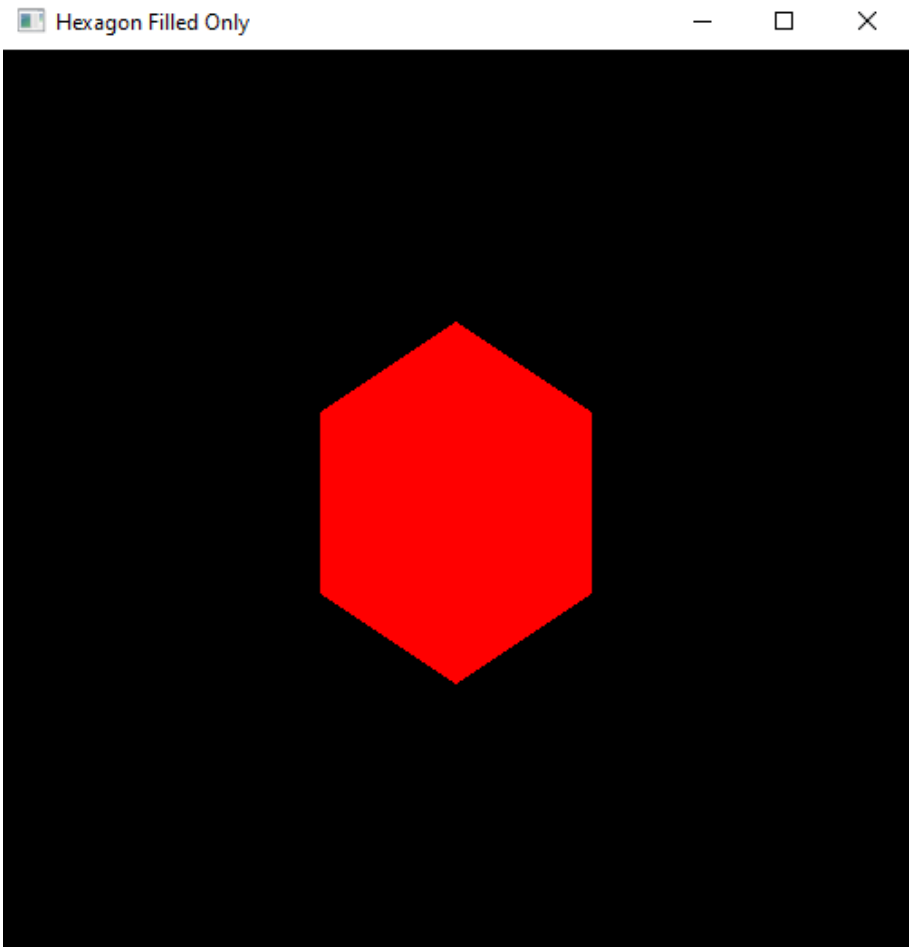
Introduction:

In this project, a hexagon is moved using translation. The shape moves with arrow key input.

Contents:

- **Functions Used:** `drawHexagon(),glTranslatef(),glBegin(),glVertex2f()`
- **Concept Used:** Translation Transformation
- **Shape Used:** Hexagon

Graph:



Draw a hexagon and show its movement in different directions.

Code (Drawing Portion Only):

```
glPushMatrix();
glTranslatef(tx, ty, 0);
drawHexagon();
glPopMatrix();
```

Discussion:

The task was to move a hexagon. I used translation and arrow keys to change the position of the shape.

Title: Reflection of a Hexagon Using OpenGL

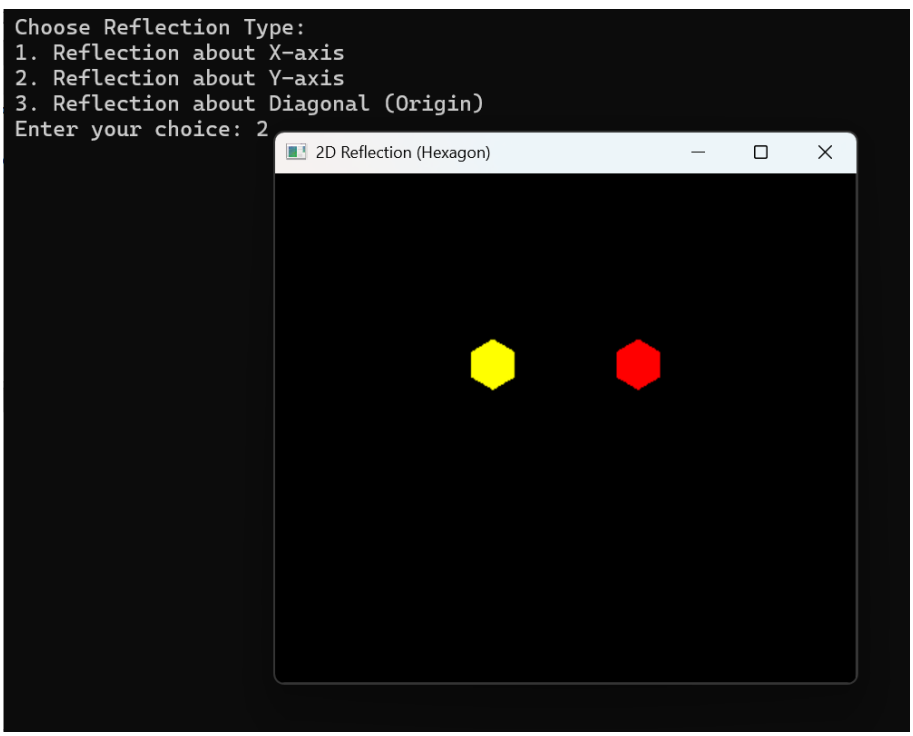
Introduction:

In this project, a hexagon is reflected using transformation. The reflected shape is shown based on user choice.

Contents:

- **Functions Used:** `drawHexagon(),glBegin(),glVertex2f(),glColor3f()`
- **Concept Used:** Reflection Transformation
- **Shape Used:** Hexagon

Graph:



Draw one hexagon and show its reflected position (X-axis / Y-axis / origin).

Code (Drawing Portion Only):

```
// Original
drawHexagon(x, y);


// Reflected
drawHexagon(rx, ry);
```

Discussion:

The task was to reflect a hexagon. I changed the coordinate signs to get reflection across X-axis, Y-axis, or origin.

Title: Shearing of a Hexagon Using OpenGL

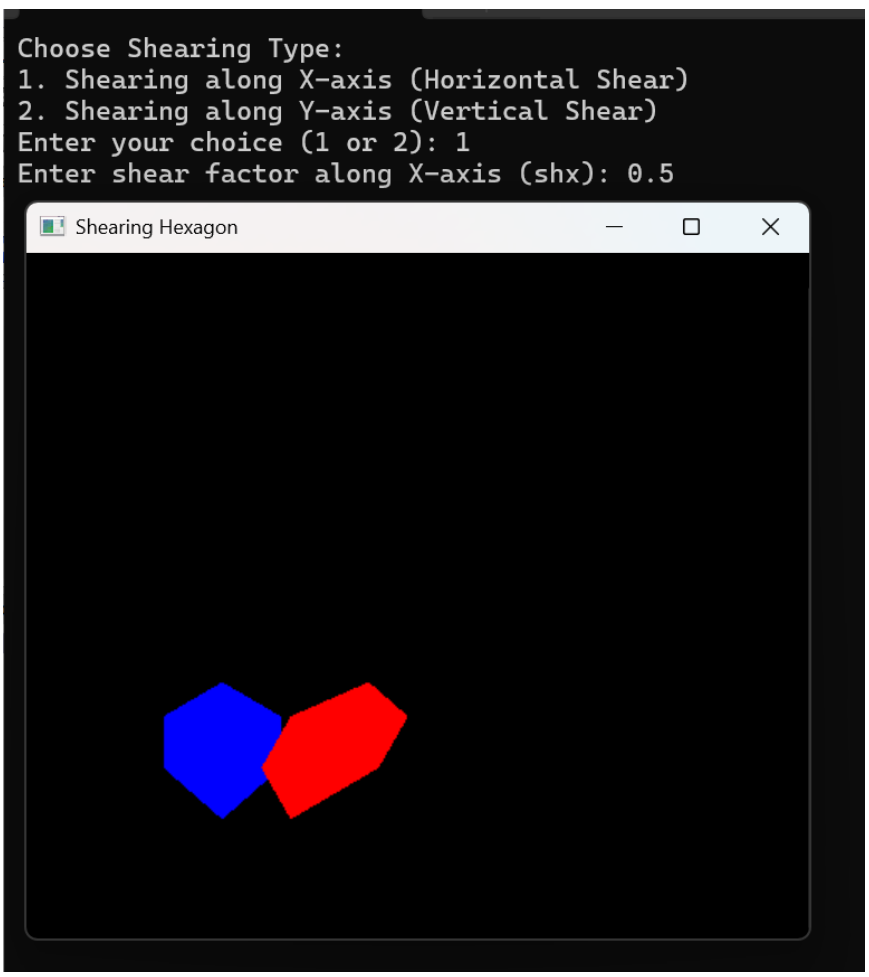
Introduction:

In this project, a hexagon is transformed using shearing. The shape is changed along X-axis or Y-axis.

Contents:

- **Functions Used:** `glBegin(),glVertex2f(),glColor3f()`
- **Concept Used:** Shearing Transformation
- **Shape Used:** Hexagon

Graph:



Draw one original hexagon and another slanted hexagon after shearing.

Code (Drawing Portion Only):

```
// Original
glBegin(GL_POLYGON);
for(int i=0;i<6;i++)
    glVertex2f(x[i], y[i]);
glEnd();

// Sheared
glBegin(GL_POLYGON);
for(int i=0;i<6;i++)
    glVertex2f(nx[i], ny[i]);
glEnd();
```

Discussion:

The task was to shear a hexagon. I changed coordinates using shear factors to tilt the shape.

Title: Drawing Animated Bangladesh Flag Using OpenGL

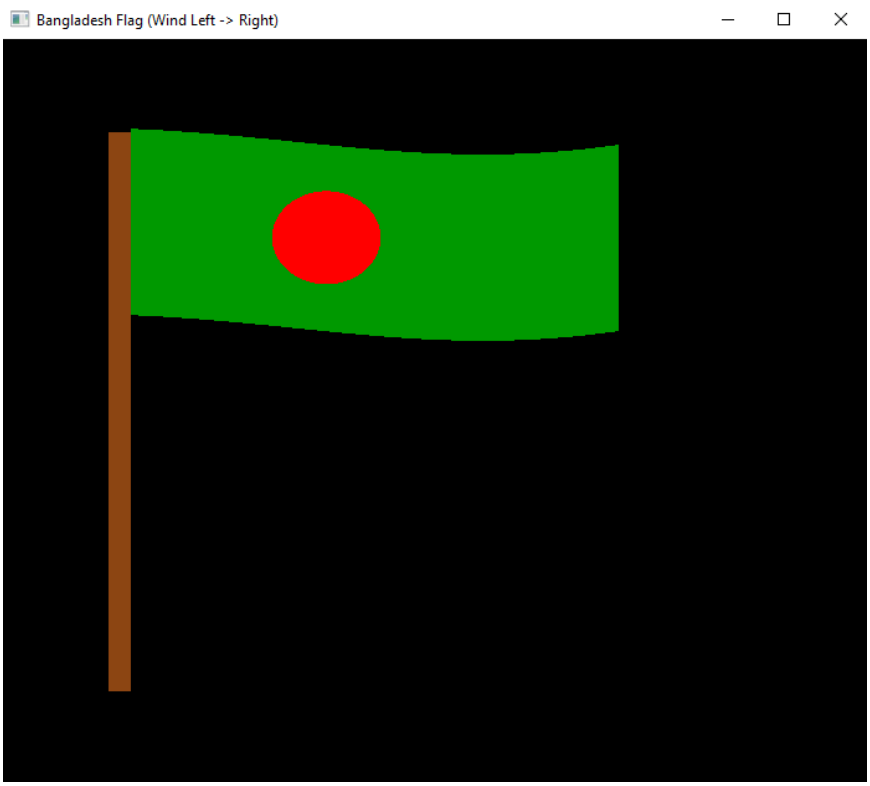
Introduction:

In this project, the Bangladesh flag is drawn with animation. The flag moves like wind from left to right.

Contents:

- **Functions Used:** `drawPole(),drawFlag(),glBegin(),glVertex2f(),glColor3f()`
- **Concept Used:** Animation (using sine wave)
- **Shape Used:** Rectangle (flag), Circle (sun), Quad (pole)

Graph:



Draw one original hexagon and another slanted hexagon after shearing.

Code (Drawing Portion Only):

```
// Original
glBegin(GL_POLYGON);
for(int i=0;i<6;i++)
    glVertex2f(x[i], y[i]);
glEnd();

// Sheared
glBegin(GL_POLYGON);
for(int i=0;i<6;i++)
    glVertex2f(nx[i], ny[i]);
glEnd();
```

Discussion:

The task was to shear a hexagon. I changed coordinates using shear factors to tilt the shape.

Title: Bangladesh Flag with All Transformations Using OpenGL

Introduction:

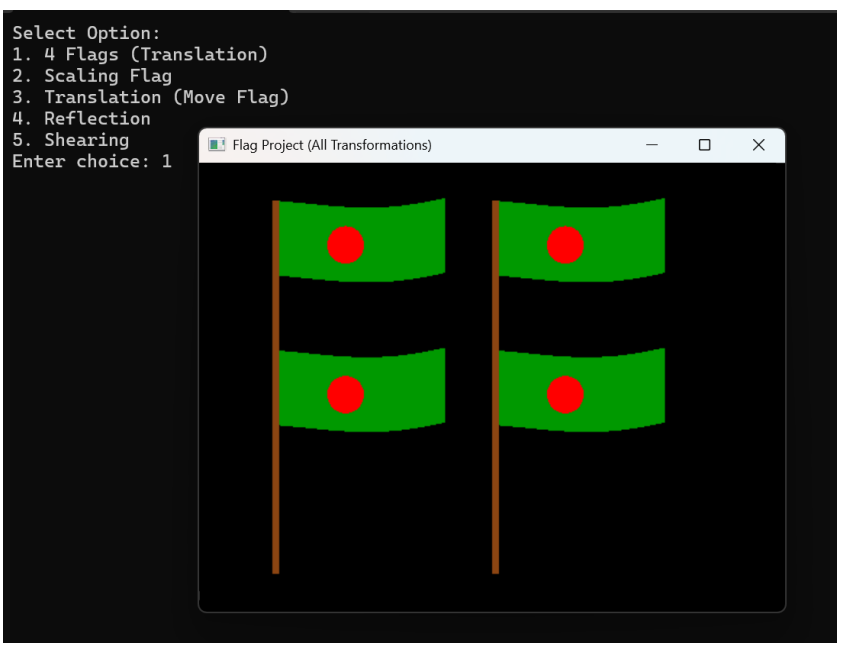
In this project, the Bangladesh flag is drawn with animation and transformations. Different operations like scaling, translation, reflection, and shearing are applied.

Contents:

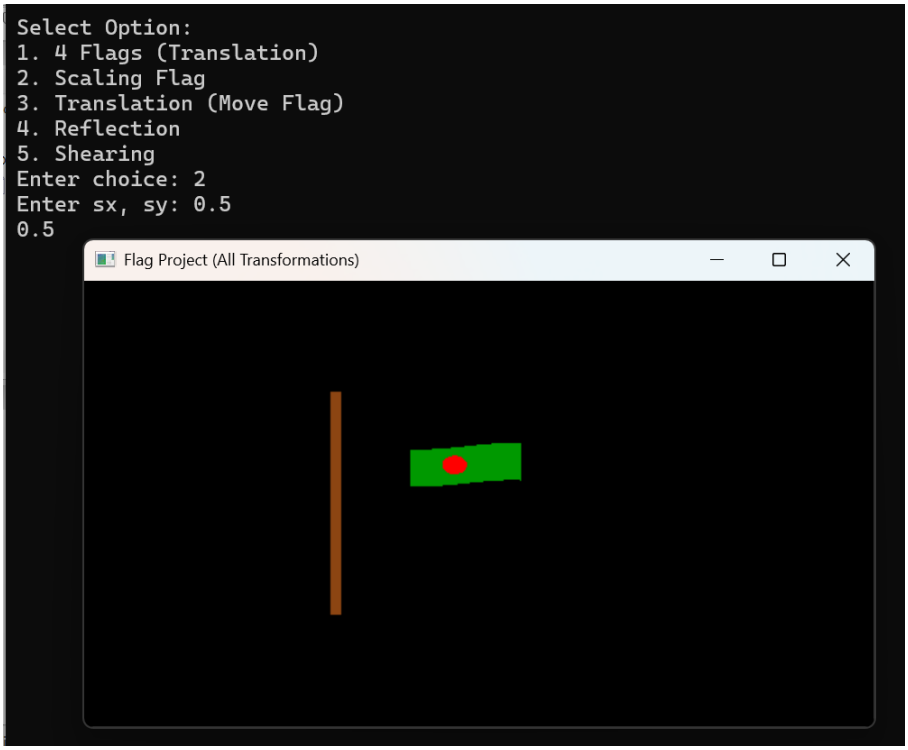
- **Functions Used:** `drawFlag()`, `drawPole()`, `glTranslatef()`, `glScalef()`, `glMultMatrixf()`
- **Concept Used:** Translation, Scaling, Reflection, Shearing, Animation
- **Shape Used:** Rectangle (flag), Circle (sun), Quad (pole)

Graph:

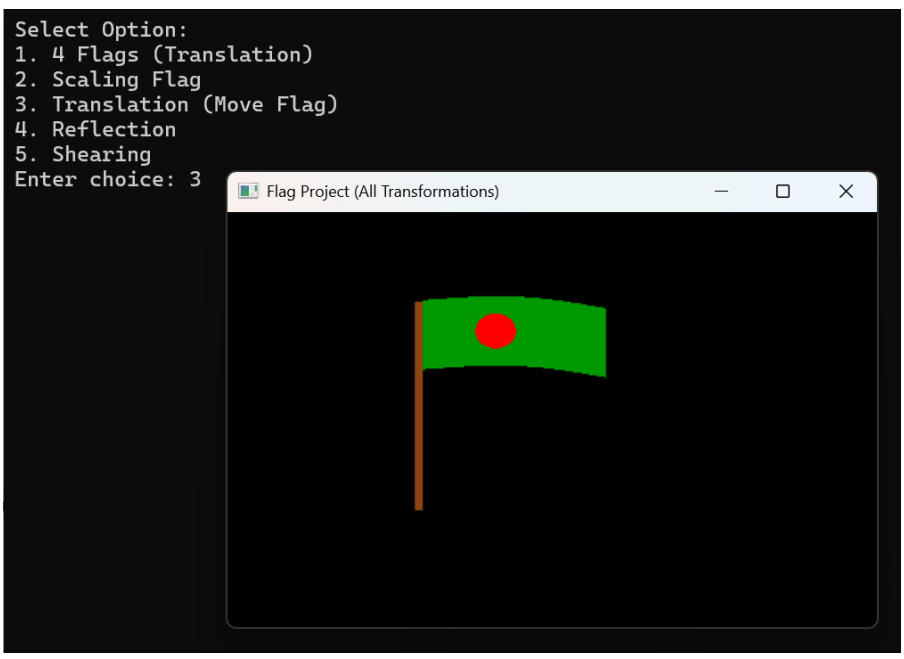
1. **4 Flags (Translation)**



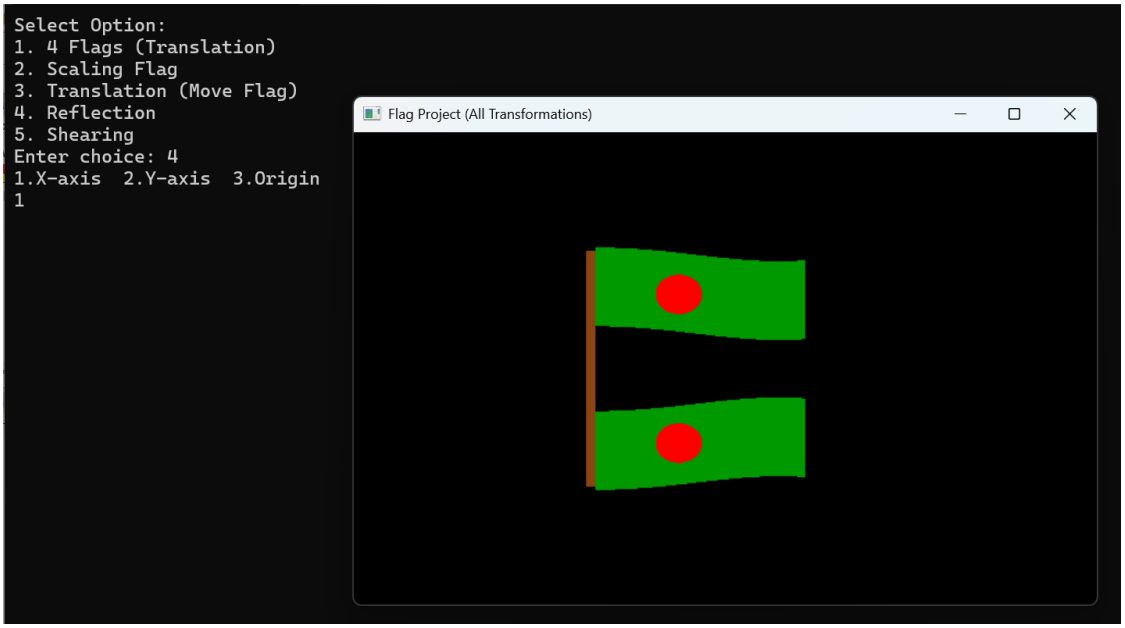
2. Scaling Flag



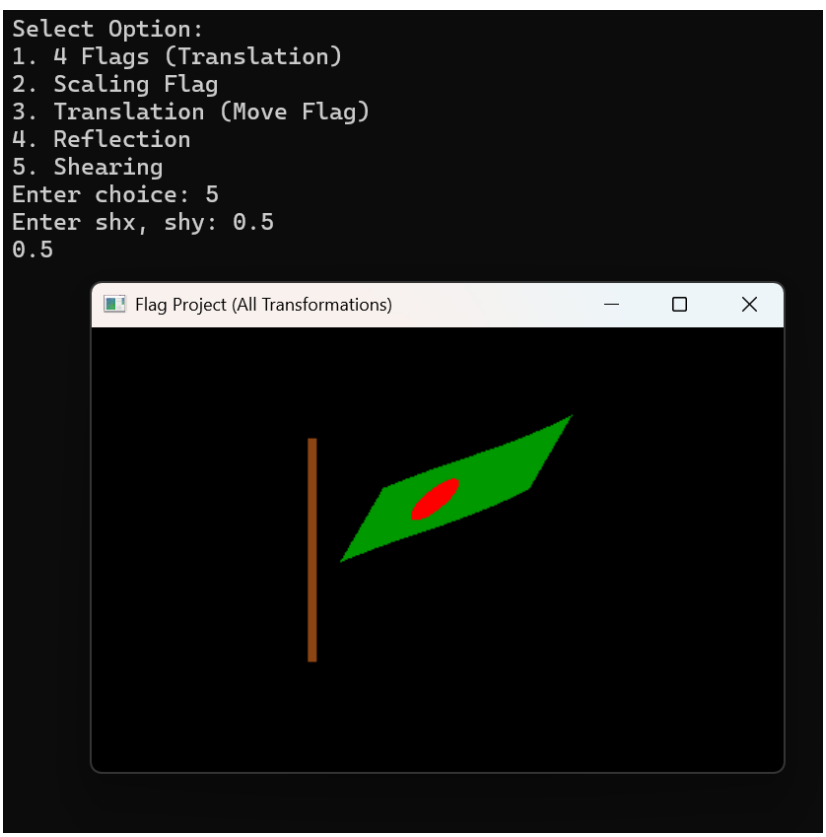
3. Translation (Move Flag)



4. Reflection



5. Shearing



Draw a flag and show different transformations like move, scale, reflect, and shear.

Full source code : Click here

Code (Drawing Portion Only):

```
// Translation
glTranslatef(tx, ty, 0);

// Scaling
glScalef(sx, sy, 1);

// Reflection
glScalef(1, -1, 1);

// Shearing
glMultMatrixf(shear);

// Draw flag
drawPole();
drawFlag();
```

Discussion:

The task was to apply different transformations on a flag. I used OpenGL transformation functions to move, resize, reflect, and shear the flag with animation.